

**SCIENTIFIC REPORT
OF
FIFTH INDIAN EXPEDITION TO ANTARCTICA**

TECHNICAL PUBLICATION No. 5



सत्यमेव जयते

**DEPARTMENT OF OCEAN DEVELOPMENT
CGO COMPLEX, LODI ROAD
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CONTENTS

	PAGE
Foreword	1
Preface	3-4
Participants	5-7
The Team	9-23
The Fifth Indian Voyage to Antarctica <i>M. K. Kaul</i>	25-53
Section-I Geology and Geophysics	
Geological set-up of a part of Central Queen Maud Land, East Antarctica <i>M. K. Kaul, S. Mukerji, R. K. Singh, D. Srivastava and S. Jayaram</i>	57-97
Anorthosites of Gruber Massif, Central Queen Maud Land, East Antarctica—An Appraisal . <i>S. Mukerji, M. K. Kaul, R. K. Singh, D. Srivastava and S. Jayaram</i>	99-108
A Short Account of the Basic and Ultrabasic Rocks Occurring between Schirmacher Hills and Gruber Massif, Central Queen Maud Land, East Antarctica . <i>R. K. Singh, S. Mukerji, S. Jayaram, D. Srivastava and M. K. Kaul</i>	109-119
An outline of the geology of the Nunataks between Eastern Wohlthat Range and Schirmacher Hills, Central Queen Maud Land, East Antarctica <i>S. Mukerji, M. K. Kaul, R. K. Singh, D. Srivastava and S. Jayaram</i>	121-133
A Note on the Petrochemical Characteristics of a Nodular Basalt Dyke from Schirmacher Hills, East Antarctica <i>M. K. Kaul, S. K. Chakraborty, R. K. Singh, S. Mukerji, D. Srivastava and S. Jayaram</i>	135-144
Geophysical Investigations in Schirmacher Landmass <i>Shikhar Ch. Jain, Ratan Dhar and K. N. S. Reddy</i>	145-150
Magnetic and Seismic Investigations on the Ice-Shelf around the Indian Permanent Station in Antarctica <i>Shikhar Ch. Jain, Ratan Dhar and K. N. S. Reddy</i>	151-155
Geotechnical Properties of the Antarctic Rocks <i>C. Mukherjee, K. Seshagiri and T. Ramamurthy</i>	157-167
Section-II Glaciology	
Oxygen Isotope Studies in Antarctica <i>V. N. Nijampurkar, S. K. Bhattacharya, S. Mukerji, R. K. Singh and D. Srivastava</i>	171-179
Polar Fallout of Radioisotopes ³² Si, ⁷ Be, ²¹⁰ Pb, ¹³⁷ Cs and ²³⁹ Pu at Dakshin Gangotri, Antarctica <i>V. N. Nijampurkar and D. K. Rao</i>	181-188

Snow Accumulation and Ablation Pattern on Ice Shelf near Dakshin Gangotri, Antarctica, and Development of fast Ice off Dakshin Gangotri	189-204
..... R. K. Singh, S. Mukerji, D. Srivastava and M. K. Kaul	
Observations on the Changes in the Snout of Dakshin Gangotri Glacier, Antarctica . M. K. Kaul, R. K. Singh, D. Srivastava, S. Mukerji and S. Jayaram	205-209
Some Observations on the Glacial Geomorphological Features of Wohlthat Mountains, Central Queen Maud Land, Antarctica	211-218
..... D. Srivastava, M. K. Kaul, R. K. Singh, S. Mukerji and S. Jayaram	
Glaciochemical Studies in Antarctica	219-224
..... V. N. Nijampurkar	
Radioactivity Studies in Antarctica	225-233
V. N. Nijampurkar, D. K. Rao, M. K. Kaul, S. Mukerji, R. K. Singh and D. Srivastava	
Glacier Inventory of Wohlthat Mountain Chain, Queen Maud Land, Antarctica	235-245
..... D. Srivastava, M. K. Kaul, R. K. Singh, S. Mukerji and S. Jayaram	
Ice Flow Conditions in the Ice Sheet Draining Part of the Central Queen Maud Land, East Antarctica	247-256
..... M. K. Kaul, R. K. Singh, D. Srivastava, S. Mukerji and S. Jayaram	

Section-III Meteorology

Meteorological Studies at Antarctica During The Period March, 1986 to February, 1987	259-281
..... T. V. P. Bhaskara Rao	
Blizzard Storms: Coastal Regions of Indian Antarctic Station, Dakshin Gangotri (1985-86 Summer)	283-287
..... A. C. Sharma	
Automatic Weather Station at Dakshin Gangotri, Antarctica	289-297
..... N. Y. Apte and T. V. P. Bhaskara Rao	
A Note on the Role of Sub-Tropical Anticyclones of Southern Hemisphere on the Severity and Persistence of Weather over Coastal Antarctica	299-301
..... A. C. Sharma	
Typical Meteorological and Oceanological Situation Encountered during the Fifth Indian Scientific Expedition to Antarctica (1985-86)	303-308
..... Surinderjit Singh	

Section-IV Biology and Oceanography

Primary and Extracellular Production during Austral Summer at 70°S, 12°E, Antarctica	311-320
..... A. Pant	
Short Term Variation in Particulate Organic Matter in the Shelf Waters of the Princess Astrid Coast, Antarctica . . V. K. Dhargalkar and N. B. Bhosle	321-330
Seasonal Variation in Particulate Organic Matter and its Constituent Fractions under the Ice covered sea near the Shelf, Antarctica	331-339
..... V. K. Dhargalkar	
Studies on the Effect of Antarctic Environment on Some Saprophytic Tropical Fungi	341-349
..... H. M. Dayal, G. P. Tandon, K. C. Tewari, Kamlesh Mehta, R. S. Pandey and H. C. Pant	

Section-V Geomagnetism and Radiophysics

A Comparative study of Quiet Day Variation of the Geomagnetic Field at Dakshin Gangotri (India) and Novolazarevskaya (USSR) in Antarctica	<i>G. K. Rangarajan and Ajay Dhar</i>	353-361
Some Interesting Features of the Infrasonic Observation carried out during the Fifth Indian Antarctic Expedition (1985-86)	<i>R. Venkatachari, A. K. Saha and D. R. Nakra</i>	363-366
Some Features of PC5 Magnetic Pulsations in Dakshin Gangotri, Antarctica	<i>G. K. Rangarajan and Ajay Dhar</i>	367-381

Section-VI Remote Sensing

Remote Sensed Data Analysis for Geology and Glacial Geology of Wohlthat Muhlig-Hofmann Mountain Chain in Queen Maud Land, East Antarctica	<i>A. Narendra Nath</i>	385-418
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Section-VII Materials Research

Clothing for Antarctica	<i>G. N. Chatterjee, R. P. Agarwal, V. K. Tandon and B. K. Banerjee</i>	421-432
Fracture and Fatigue Studies on Mild and Structural Steels For Use in Antarctica	<i>Sameer Vijayakar</i>	433-441

Section-VIII Non-Conventional Energy

Experiments on Harnessing Non-Conventional Energy Sources in Antarctica	<i>K. C. Ramakrishnan</i>	445-453
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Section-IX General

Mineralising Snow-Melt Water	<i>B. K. Banerjee</i>	457-464
A Note on Comparative Methane Measurements on an Antarctic Air Sample	<i>D. C. Parashar, J. Rai, Prabhat K. Gupta, V. Raman, R. C. Sharma, N. Singh, H. Singh and A. K. Saha</i>	465-466
Communication System at Dakshin Gangotri, Antarctica	<i>Tata Sudhakar</i>	467-472
Prospects of Horticulture in Antarctica	<i>M. C. Joshi and B. K. Banerjee</i>	473-487

FOREWORD

The Fifth Indian Scientific Expedition to Antarctica has been another step forward in the Indian Antarctic programme. It extended its scientific activities in areas far deeper than those attempted during the earlier expeditions. Successful accomplishment of the multidisciplinary scientific work during this expedition in the Wohlthat mountains has laid a firm foundation for further expansion of the scientific activities around this mountain chain, in addition to those going on at Dakshin Gangotri and Maitree.

During the Indian Expeditions to Antarctica, both fundamental and applied researches have been carried out. Thus the rapidly advancing polar science has made spectacular progress in the country. Continuation of the work in Antarctica has led to further refinements in our knowledge and in the rediscovery of some of the salient scientific features about this continent. Our main interest in Antarctica has been in this rediscovery so as to establish further landmarks in Indian science.

The Fifth expedition team consisted of 88 members. It brought back a 13 member Second Wintering Team and left behind the Third Wintering Team consisting of 14 members at the Dakshin Gangotri station for the continuation of scientific observations and maintenance of the station.

This report includes investigations on a variety of scientific disciplines such as geology, glaciology, marine biology, meteorology, geophysics, radiophysics, geomagnetism, use of non-conventional sources of energy, remote sensing etc. These papers are indeed testimony of India's unfailing involvement in polar science.

I take this opportunity to give my highest appreciation to all those who participated in this expedition and carried out both scientific and logistic work with great dedication and confidence.

I would also like to thank Mr. M.K. Kaul, the Leader of the Fifth Expedition and his colleagues for finalising all the scientific works worthy of inclusion in this report, and to my colleague, Mr. V. Ravindranathan for taking up the responsibility of the editorial work of this report and for meticulously going through each paper.

I am sure this report will also be well received by the Antarctic scientists throughout the world.

October 1988

S.Z. QASIM
Secretary
Department of Ocean Development
New Delhi.

PREFACE

I would like to record, first of all, with appreciation the role played by the earlier Indian expeditions to Antarctica, in building up a sound infrastructure for continued scientific activity at Antarctica. Thus the 88 member Fifth Indian Expedition to Antarctica left for the continent with the confidence that the latter was neither distant nor desolate. There was a thirteen member party of the Second Wintering already waiting to receive the expedition with the Indian tricolour fluttering at the station building at Dakshin Gangotri. The inherent hazards of Antarctica and its weather notwithstanding, the expedition was able to start the work immediately after its arrival and to stay on the continent for a much longer period, thus not only accomplishing the objectives but far exceeding its targets.

The expedition has been a fine example of co-ordination and cohesion. The expedition members, a highly motivated group, surmounted several difficulties and heartily came to each other's help in the execution of their tasks, which contributed not only to the building up of human relations but also the scientific temper. The expedition started with a note of mutual understanding and ended too on that note.

Behind the success of this expedition there was an inspiring support, encouragement and guidance received from various sources by the leader and the members, before, during, and after the expedition. On my behalf and on behalf of the members of the team I express my gratitude to all of them.

My grateful thanks are due to Shri Shivraj Patil and Shri K.R. Narayanan, Ministers of State for Science and Technology for their keen interest in this programme. To Dr. S.Z. Qasim, Secretary, Department of Ocean Development, whose deep involvement with Antarctica has established a milestone in Indian science, I am thankful for guidance and constant encouragement at every stage. Thanks are also due to Dr. S. Vardarajan, Adviser, Planning Commission, Dr. V.S. Arunachalam, Scientific Adviser to Defence Ministry, Dr. A.P. Mitra, Director General, C.S.I.R. and Shri B.K. Rao, Secretary, Department of Mines for their interest in formulation and finalisation of the expedition objectives and for their advice.

I would like to thank Shri I.G. Jhingran, the then Joint Secretary, and Shri A.K. Mathur, Financial Adviser, Department of Ocean Development (D.O.D), for their continued administrative support before and during the preparations of the expedition. I am thankful to Dr. S.N. Dwivedi Additional Secretary and Shri M.M.K. Sardana, Joint Secretary, D.O.D, for useful suggestions. My sincere thanks are due to Shri Pavan Raina, Director, D.O.D, who, by his hard work and dedication, solved many problems during the planning stage of the expedition. Dr. S.A.H. Abidi, Director, D.O.D. was always helpful, whenever approached. My thanks are due to Sq. Ldr. V.P. Savant who maintained a constant link with

the members of the team and their homes and meticulously carried out the onerous task of conveying the messages to and fro, from his desk at D.O.D. I also thank all the members of OD-I Section at D.O.D. who did a lot of work during the pre-expedition preparations and on arrival of the team in Goa.

I am thankful to Shri S.K. Mukherji, former-Director General and Shri D.P. Dhoundial, Director General, Geological Survey of India for their keen interest and support, and to Shri A.P. Tewari, Deputy Director General, Shri V.K. Raina, Deputy Director General, and Shri M.L. Parimoo, Director, Geological Survey of India, for their guidance. I would like to thank Director General, India Meteorological Department; Director, National Geophysical Research Institute; Director, National Institute of Oceanography; Director, Bharat Heavy Electricals Limited; Chairman, Oil & Natural Gas Commission; Director, Indian Institute of Geomagnetism; Director, National Physical Laboratory; Director, Physical Research Laboratory; Director, National Remote Sensing Agency; Director, D.M.S.R.D.E. for their constant support in the execution of the scientific programme.

I am thankful to the Chief of the Army Staff, Chief of the Naval Staff and Chief of the Air Staff for providing a dedicated group of officers and men for the expedition. I am also thankful to Maj. Gen. R.N. Verma; Vice Admiral L. Ramdas; Air Marshal S. Raghavendran; Director, C.M.E., Pune; Maj. Gen. R.A. Kadambi; Brig. Lalji D. Singh; Capt. P.I. Oommen; Air Commodore Shankaran; Col. D.N. Tankha, for their active involvement in the logistics of the expedition. Thanks are due to Chief Producer, Films Division for his cooperation. I would like to express my gratitude to S/Shri A.K. Raina and R.N. Sood of the Liaison Office, G.S.I., New Delhi, for their generous help at every stage.

It is a pleasure to acknowledge the help rendered by Goa Shipyard, Goa Port Authority and M/s Hira Lal & Co. before sailing from and after the arrival of the team in Goa. The Captain and crew of the ship M.V. Thuleland were ever helpful and always co-operative.

I would personally like to thank all the members of the expedition, respective team leaders and members of various committees framed during the expedition, for their valuable suggestions and active co-operation which made the expedition a success.

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Leader**

October 1988

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THE TEAM



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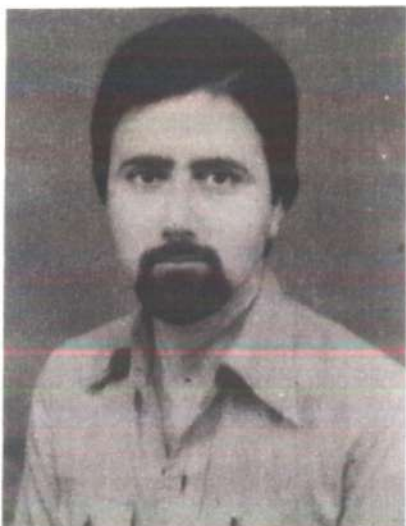
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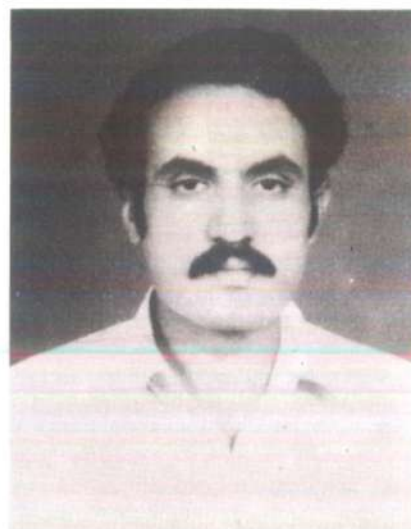
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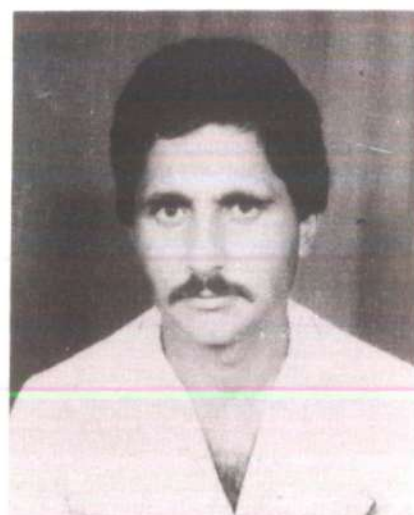
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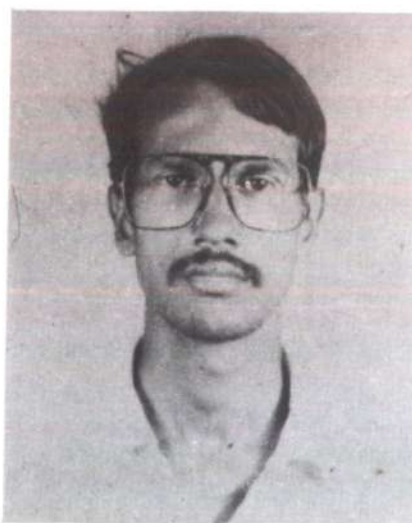
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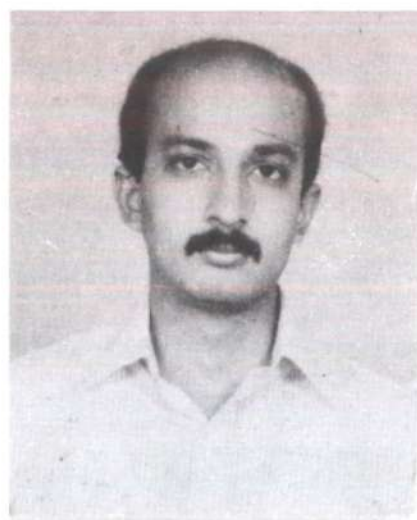
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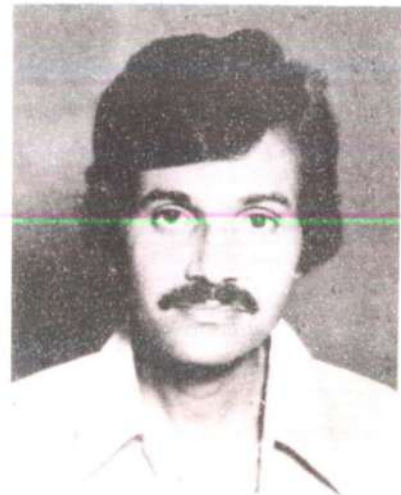
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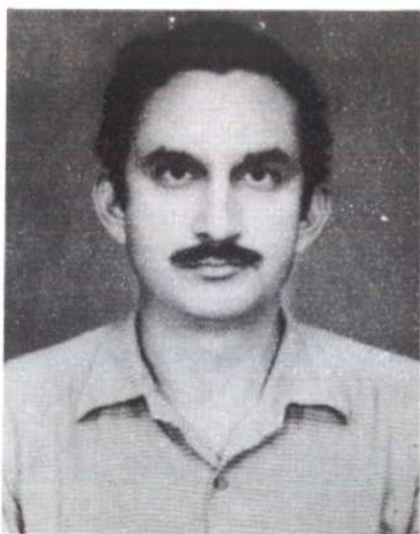
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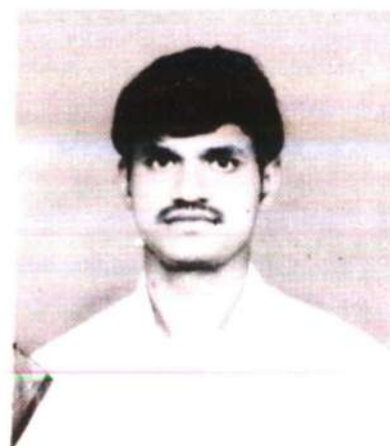
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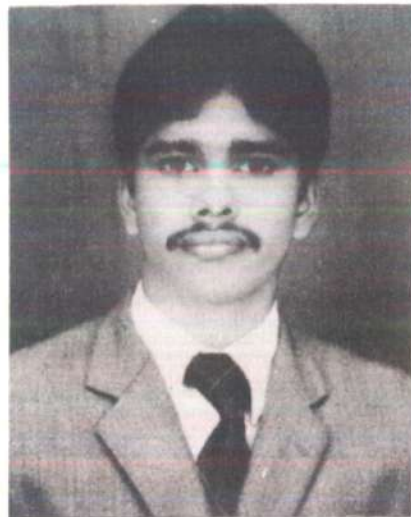
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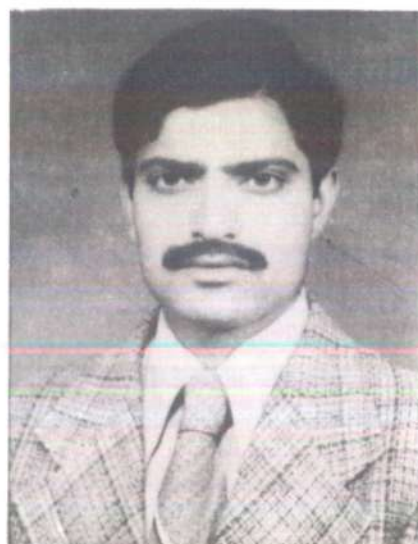
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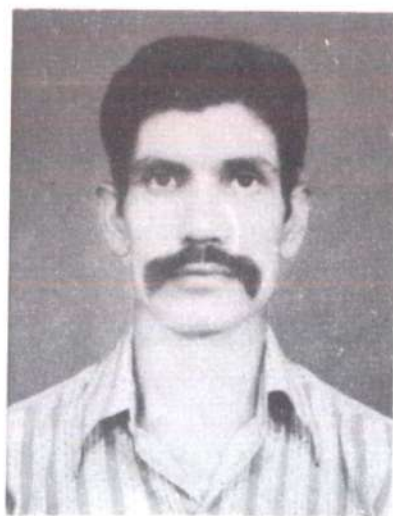
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